

ASHUTOSH ROY

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EDUCATION

Chhattisgarh Swami Vivekananda Technical University, Bhilai

2022 – 2026

B.Tech in Computer Science and Engineering (Artificial Intelligence)

PROJECTS

Multitask Speech Emotion & Stress Detection | Python, PyTorch, Hugging Face Transformers, NumPy **Present**

- Built a hybrid deep learning model (Wav2Vec2 + BiLSTM) for multi-task emotion classification and stress prediction.
- Utilized 3 datasets (RAVDESS, TESS, SAVEE) with a unified preprocessing pipeline across 5 emotion classes.
- Achieved 85.9% accuracy, 0.856 F1-score, and RMSE of 0.1268 through end-to-end pipeline implementation.

Exam-Helper RAG | Python, LLaMA3, FAISS/Pinecone, LangChain, RAG

Dec 2024 – April 2025

- Built an LLM-powered study assistant using LLaMA 3 with RAG architecture for contextual question answering.
- Implemented efficient retrieval using FAISS/Pinecone to fetch relevant data from large document sets.
- Developed multimodal querying system supporting PDFs, images, and YouTube-based content.

Medical Chatbot (LLM-based) | Python, QLoRA, OCR (Tesseract), Transformers

Mar 2024 — Apr 2024

- Fine-tuned Mistral 7B using QLoRA for domain-specific medical text understanding.
- Built an OCR pipeline to extract and process information from medical reports.
- Improved data preprocessing and structuring for better model inference and response accuracy.

Music Identification App | Python, ACRCLOUD API, Streamlit, NumPy, Requests

Sep 2024 — Dec 2024

- Built a real-time audio recognition system using ACRCLOUD API for song identification.
- Designed an end-to-end pipeline from audio input to feature extraction and matching.
- Retrieved song metadata by integrating external APIs for accurate identification results.

EXPERIENCE

Indian Institute of Technology (IIT) Delhi

Jan 2025 – June 2025

Research Intern

- Developed OCR-powered document understanding pipeline integrated with RAG for archaeological datasets.
- Implemented hybrid search using FAISS/Pinecone and Apache Solr (semantic + keyword retrieval).
- Benchmarked OCR systems (Tesseract, Marker, Gemini OCR), improving extraction accuracy.
- Built scalable APIs using FastAPI for search, retrieval, and LLM-based querying.

PUBLICATIONS

SARCH: Multimodal Search for Archaeological Archives ↗

- Co-authored with IIT Delhi.
- Developed multimodal retrieval system using OCR + image understanding + hybrid search.

Multitask Speech Emotion and Stress Detection Using Wav2Vec2-BiLSTM

- Accepted at RECCAP 2026, IIT Palakkad.
- Focused on speech-based emotion classification and stress prediction using a hybrid deep learning framework.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL

AI/Machine Learning: Deep Learning, Machine Learning, Computer Vision, Transformers, RAG, LLMs

Libraries & Frameworks: PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, Matplotlib, Seaborn, PyQt5, Streamlit, TensorFlow, Hugging Face

Data Analysis & Visualization: Excel, Power BI, Tableau

Databases: MySQL

Tools & Platforms: Git, GitHub, Linux, VS Code, PyCharm, MATLAB, Jupyter Notebook, Google Colab, FastAPI, Docker, Streamlit, FAISS, Pinecone

ACHIEVEMENTS

- First Runner-up** – Data Analytics Hackathon, IIM Nagpur
- Strong problem-solving foundation** with 100+ DSA problems solved across core patterns
- EF SET English Certificate**–71/100 (C2 Proficient), Issued Jun 2025